Total No. of Questions : 8]

SEAT No. :

PB3844

[Total No. of Pages : 3

[6262]-106

T.E.(Electronics &Computer Engineering)

DATABASE MANAGEMENT SYSTEMS

(2019 Pattern) (Semester -I) (310341)

Time : 2½ Hours] ***[Max. Marks :70***

Instructions to the candidates:

1) *Solve Q.1or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right side indicate full marks.*
3) *Assume suitable data, if neccessary.*

Q1) a) Explain PL/SQL stored procedures with syntax.Compare PL/SQL stored procedure and stored function. **[6]**

b) Consider the following schema- **[6]**
student (rollno, name, class, fees_paid,)
Write SQL queries for following requirements-
i) Retrieve all records from the student table.
ii) Retrieve rollno and name from the student table.
iii) Retrieve the rollno and names of students whose fees_paid is less than 30000.
iv) Delete all records from the student table who have paid the fees greater than30000.

c) Explain any four types of join operations in SQL with example. **[8]**

OR

Q2) a) Explain any six aggregate functions in SQL by giving suitable example. **[6]**

b) Explain SQL ORDER BY and WHERE Clause with example. **[6]**

c) Explain the following SQL DML statements by giving suitable example. **[8]**
i) SELECT
ii) INSERT
iii) UPDATE
iv) DELETE

Q3) a) Explain concept of schedule in DBMS transaction in detail.with the help of transaction states diagram explain the different transaction states in DBMS during its execution. **[8]**

b) Explain the need of concurreny control in DBMS. Explain time stamp based concurrency control. **[8]**

OR

Q4) a) Explain the concept of serializable schedule. Explain veiw serializable schedule with example. Compare conflict and view serializability. **[8]**

b) Check whether the given non-serial schedule is conflict serializable or not. If serializable, find conflict equivalent schedule. **[8]**

T1	T2	T3
	R(X)	
		R(X)
W(Y)		
	W(X)	
		R(Y)
	W(Y)	

Q5) a) Explain need and goal of parallel databases. Draw and explain two types of parallel database architectures. **[8]**

b) Explain data replication and data fragmentation in distributed data storage. **[8]**

OR

Q6) a) What are commit protocols in distributed DBMS? Explain two phase commit protocol in detail with necessary illustration. **[8]**

b) What is parallel databases? Explain performance parameters for parallel databases. **[8]**

Q7) a) Explain structured, unstructured, and semi-structured data and provide examples for each type. **[9]**

b) Discuss the BASE properties of NoSQL dastabases. Compare ACID and BASE properties. Write advantages and disadvantages of BASE properties. **[9]**

OR

Q8) a) Expalin in short following Types of NoSQL Databases **[9]**

i) Key-value Pair Based

ii) Column-oriented

iii) Document-oriented

b) Explain the syntax and usage of CRUD operations (Create, Read, Update, Delete) in MongoDB with suitable examples. **[9]**

Total No. of Questions : 8]

SEAT No. :

PA-1593

[Total No. of Pages : 2

[5926]-218

T.E. (Electronics and Computer Engineering)
DATABASE MANAGEMENT SYSTEMS
(2019 Pattern) (Semester-I) (310341)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Assume suitable data, if necessary.***

Q1) a) Compare DELETE and TRUNCATE queries in SQL. **[6]**

b) Explain any six numeric functions in SQL. **[6]**

c) What is SQL? Explain CREATE, ALTER and DROP queries with proper syntax. **[8]**

OR

Q2) a) What is mean by constraint in SQL? Explain any two constraints with suitable example. **[6]**

b) Compare stored functions and stored procedure in SQL. **[6]**

c) Explain the following set operations. **[8]**

i) UNION

ii) UNION ALL

iii) INTERSECT

iv) MINUS

Q3) a) Explain concept of Schedule. With the help of transaction state diagram Explain different types of states in DBMS transactions. **[8]**

b) Explain need of concurrency control in DBMS. Explain lock based concurrency control protocol. **[8]**

OR

Q4) a) Explain concept of serializable schedule with example. Explain conflicts serializable and view serializable schedule. **[8]**

b) With the help of neat diagram explain deadlock in DBMS. Explain deadlock detection and deadlock prevention. **[8]**

Q5) a) Explain the parameters throughput, response time, speed-up and scale-up related to parallel database. **[8]**

b) Explain two tier and three tier database architecture. Compare two tier and three tier database architecture. **[8]**

OR

Q6) a) With the help of diagram explain distributed database system. List features of distributed database. State the advantages of distributed databases. **[8]**

b) What are the design considerations for distributed databases? Explain. **[8]**

Q7) a) Explain the difference between SQL and NoSQL. **[6]**

b) Explain MapReduce operation in MongoDB with suitable example. **[6]**

c) Explain CREATE, READ and UPDATE in MongoDB with examples. **[6]**

OR

Q8) a) Explain any three aggregation functions using MongoDB with suitable example. **[6]**

b) List different NoSQL data Models and explain Document based Data Model. **[6]**

c) Enlist CRUD operations in MongoDB database with syntax. **[6]**

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Total No. of Questions : 8]

SEAT No. :

P7603

[Total No. of Pages : 2

[6180]-122

T.E. (Electronics and Computer Engineering)

DATABASE MANAGEMENT SYSTEM

(2019 Pattern) (Semester - I) (310341)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right side indicate full marks.***
3) ***Assume suitable data, if necessary.***

Q1) a) Explain the various clauses for SELECT Query with example. **[6]**

b) Write a short note Aggregate Functions. **[6]**

c) Explain stored procedure and stored function. Differentiate between stored procedure and stored function. **[8]**

OR

Q2) a) Explain pattern matching operator "LIKE" **[6]**

b) Explain any two join operation with example. **[6]**

c) Define Views. Explain Create, Update and Delete Views with examples. **[8]**

Q3) a) What is serializability? Explain conflict serializability and view serializabilty with example. **[8]**

b) Explain in detail ACID properties of transactions. **[8]**

OR

Q4) a) What is a deadlock? Explain how deadlock detection and prevention is done. **[8]**

b) Explain the need of a concurrency control system? How is it achieved with timestamp based protocol? **[8]**

Q5) a) Explain two important issues Speed Up and Scale Up in case of parallel databases. **[8]**

b) Define a distributed database. Explain advantages and disadvantages of distributed database. **[8]**

OR

Q6) a) Enlist different Parallel database architectures. Explain any two in detail. **[8]**

b) Describe concurrency control in distributed databases. **[8]**

Q7) a) Explain MapReduce operation in MongoDB with suitable example. **[6]**

b) Explain ACID VS BASE properties. **[6]**

c) List difference between RDBMS and No SQL. **[6]**

OR

Q8) a) Explain any three aggregation functions using MongoDB with suitable example. **[6]**

b) List different NO SQL data Models and explain Document Based Data Model. **[6]**

c) Explain CREATE, READ and UPDATE in MongoDB with examples. **[6]**